

Conclusion

We presented MMRL++, a parameter-efficient approach to multi-modal representation learning for vision-language models. Our method introduces two key innovations: Shared-Residual Representation Aligner (SRRA), which reduces trainable parameters by over 80% through shared base weights and layer-specific low-rank residuals, and Progressive Representation Composition (PRC), which enables inter-layer semantic flow through beta-weighted composition. Extensive experiments across three evaluation settings (base-to-novel generalization, few-shot learning, cross-dataset transfer) and 15 datasets demonstrate that MMRL++ achieves superior parameter-performance trade-offs compared to prior methods including MMRL and MaPLe.

Our ablation studies validate that both SRRA and PRC contribute independently to performance gains. SRRA’s shared base weights enable gradient sharing across layers, improving training stability and convergence, while PRC’s progressive composition creates structured semantic flow that enhances representation quality. The combination of these mechanisms addresses fundamental limitations of independent per-layer projections used in prior work.

The parameter efficiency of MMRL++ has practical implications for deployment in resource-constrained scenarios. By reducing trainable parameters while maintaining or improving performance, our method enables vision-language adaptation on edge devices, mobile platforms, and embedded systems. This democratizes access to powerful vision-language capabilities and reduces the environmental impact of fine-tuning through lower energy consumption.

Future work could explore several directions. First, adaptive or learnable beta parameters could eliminate the need for dataset-specific tuning. Second, extending MMRL++ to very deep networks (>24 layers) through hybrid SRRA-independent projection architectures could further improve scalability. Third, incorporating domain adaptation techniques could enhance cross-dataset transfer performance. Finally, applying MMRL++ to other vision-language architectures (BLIP, ALBEF) and modalities (video, audio) would broaden its applicability.

In conclusion, MMRL++ demonstrates that parameter efficiency and performance are not mutually exclusive in vision-language adaptation. Through careful architectural design that enables gradient sharing and inter-layer composition, we achieve state-of-the-art results with substantially fewer trainable parameters, paving the way for more efficient and accessible vision-language model deployment.